Lab Presentation Assessment

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	5
Delivery	2					
Content/Organization	2					
Enthusiasm/Audience Awareness	1					
Total obtained mark						
Comments						

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Team Information:

Name: Farhan Ibn Mustafa

ID: 0242310005341325

Name: Yeasmin Akter

ID: 0242310005341207

Name: Md Mobin

ID: 0242310005341226

Batch: 40th

Section: A2

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Course Teacher Name: Mr. Humaun Kabir Nayem

Designation: Lecturer

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City Water Distribution and Management System

Introduction

In today's cities, it's very important to manage water properly so that clean and safe water reaches every home and business. The **City Water Distribution and Management System** helps make this easier. It keeps track of how much water people use, checks the water quality, and quickly handles any problems like leaks. This system is useful for both city workers and customers. It helps things run more smoothly, keeps people happy, and supports smart, sustainable water use.

Objective

The primary objective of the City Water Distribution and Management System is to provide an integrated, user-friendly platform that supports the smooth operation of water delivery, usage tracking, billing, and issue resolution. The system aims to:

- Ensure clean water delivery to residents and businesses.
- Monitor water usage and manage billing efficiently.
- Detect and resolve issues such as leaks and water quality concerns quickly.
- Promote water conservation and sustainable usage.

Scope

The system provides water management for the city, focusing on the following features:

- Water delivery and distribution management.
- Water quality monitoring.
- Water usage tracking and billing.

- Maintenance and issue reporting.
- Notifications and alerts for excessive usage or system faults.

Key Features

1. **User Registration and Login**
   The system allows customers and admin users to register securely and log in with authentication mechanisms in place, ensuring that only authorized users can access the system.

2. **Water Usage Tracking**
   Customers are provided with water meters to track their consumption. The system automatically records usage and provides monthly reports to customers.

3. **Water Quality Monitoring**
   The system continuously monitors water quality to ensure it meets health standards. If any irregularities are detected, the system alerts the admin immediately.

4. **Billing and Payment**
   Based on the recorded water usage, the system generates bills and allows customers to pay online or in person through various payment options.

5. **Issue Reporting**
   Customers can report issues such as no water or leaks. The repair team receives instant notifications, enabling them to address the problem promptly.

6. **Excessive Usage Warning**
   If a customer's water usage is unusually high, the system sends a warning to both the customer and the admin, prompting an investigation.

7. **Notifications and Alerts**
   Automated notifications are sent to both customers and admins to keep them updated on billing, water usage, and maintenance issues.

8. **Helpline Support**
   A 24/7 helpline is available to assist customers with any issues or questions regarding their water service.

Scenario Writing

Scenario writing is a helpful technique for demonstrating how users interact with a system and how the system responds to these actions. Below are some scenarios that describe common use cases for the City Water Distribution and Management System:

Scenario-1: Customer Registration and Profile Creation

Scenario Description: A new customer wants to use the water distribution system.

- The customer provides their name, address, contact, and billing information.
- The system checks the information and creates a new customer account.
- The customer logs in using their new credentials.
- The system confirms the profile is created and sends a welcome notification.

Scenario-2: Water Usage Tracking

Scenario Description:

- The system tracks the customer's water usage based on data from their water meter.
- Monthly reports are generated showing total water usage.
- The system alerts the customer if their usage exceeds typical levels.

Scenario-3: Billing and Payment

Scenario Description:

- Based on the water usage data, the system generates a monthly bill.
- The customer is notified of their bill, which includes water usage, taxes, and any extra charges.
- The customer selects a payment method (online or in person) and completes the payment.
- The payment status is updated in the system.

Scenario-4: Issue Reporting

Scenario Description:

- A customer notices that their water supply is interrupted or that there is a leak.
- The customer reports the issue via the app or website.
- The repair team is notified and dispatched to resolve the issue.
- The customer receives updates on the status of the repair.

Scenario-5: Excessive Water Usage Alert

Scenario Description:

- The system detects that a customer's water usage is unusually high.
- A warning is sent to the customer, suggesting they check for potential leaks or overuse.
- The admin is also notified to monitor the situation.

Stakeholders

The key stakeholders involved in the City Water Distribution and Management System include:

1. Admin (Manager)
    - Manages the entire water distribution system.
    - Ensures that water quality is maintained, usage is monitored, and issues are resolved promptly.
    - Oversees billing and payment processes.
2. Customer (Resident or Business)
    - Uses the water provided by the system.
    - Can track water usage, report issues, and make payments.
    - Receives notifications about billing, usage, and maintenance.
3. Repair Team
    - Responds to reported issues such as leaks, interruptions, or water quality concerns.
    - Receives notifications from the system when a problem is reported by a customer.

This City Water Distribution and Management System aims to streamline water delivery and usage management, improve operational efficiency, and ensure that water-related issues are quickly resolved, ultimately leading to a better customer experience and more sustainable water use in the city.

System Design Diagrams

1. Use Case Diagram

Actors:

- Customer
- Admin (Manager)

- Repair Team

Use Cases:

- Customer Registration & Login
- View Water Usage
- View Water Quality
- Pay Water Bill
- Recharge Water Card
- Report Water Issues
- Receive Alerts & Notifications
- Admin Login & Manage Customers
- Monitor Alerts & Water Usage Reports
- View Payment History

Purpose: To display the interactions between users (customers, admin, repair team) and the system in a visual format.

2. Data Flow Diagram (DFD)

Level 0 – Context Diagram: Shows the entire system as a single process.

External Entities: Customer, Admin, Repair Team

Major Data Flows: Registration info, usage data, payments, issue reports, water quality info

Level 1 – Breakdown:

- Registration & Login Management
- Water Usage Monitoring (from Smart Meters to Database)
- Billing & Payment Processing
- Water Quality Monitoring

- Issue Reporting & Repair Dispatch
- Admin Management & Alerts Handling

Purpose: To illustrate how data is collected, processed, and shared within the system.

3. Activity Diagram

Processes:

- Customer Registration and Login
- Bill Payment Process
- Reporting an Issue and Tracking Repair
- Water Usage Tracking
- Water Quality Check and Notification
- Admin Review

Purpose: To show the sequence of steps and decisions in key user and system activities.

4. Sequence Diagram

Interactions:

- Customer registers → system verifies & stores → customer logged in
- Smart meter sends usage data → system calculates bill → sends to customer
- Customer pays → payment verified → system updates record
- Issue reported → repair team notified → status updates sent
- Admin requests reports → system generates and shows report

Purpose: To show the time-based flow of interactions between system actors and the system itself.

5. Class Diagram

Classes:

- Customer (ID, Name, Address, Usage, Bill)

- Admin (Login info, Authority Level)
- SmartMeter (MeterID, CustomerID, UsageData)
- Payment (BillID, Amount, Status, Date)
- IssueReport (ReportID, CustomerID, Description, Status)
- WaterQualityReport (ReportID, QualityMetrics, Timestamp)
- Notification (To, Message, Type, Time)
- RepairTeam (TeamID, AssignedTasks)

Purpose: To outline system structure, attributes, and relationships between classes.

6. Entity-Relationship (ER) Diagram

Entities:

- Customer
- Admin
- WaterMeter
- Bill
- Payment
- RepairRequest
- RepairTeam
- WaterQualityReport

Relationships:

- A Customer has a SmartMeter
- A Customer generates Bills
- A Bill is linked to a Payment
- A Customer reports a RepairRequest
- A RepairRequest is handled by a RepairTeam
- System generates WaterQualityReport

- Admin monitors usage and manages customers

Purpose: To design the database structure and understand entity relationships.

Expected Outcomes

- A detailed, well-documented City Water Distribution & Management System design
- Accurate water usage tracking with billing integration
- Improved customer satisfaction through alerts and updates
- Faster issue resolution with real-time repair reporting
- Centralized control and visibility for city administrators

Conclusion

The City Water Distribution and Management System is built to ensure clean, efficient, and reliable water supply for all citizens. Through automated monitoring, customer-focused services, and admin controls, it promotes operational efficiency, sustainability, and accountability in urban water management. This proposal lays the groundwork for deploying a real-time, scalable solution to modern water supply challenges.

Software Requirements Specification

FR01	**Registration**
Description	Customers and admins must register in the system to start using the service
Stakeholder	Admin, Customer

FR02	**Login**
Description	Admins and customers must log in to access their accounts.

Stakeholder	Admin, Customer

FR03	**Water Metering and Usage Tracking**
Description	Each customer will have a water meter to track their usage. The system will record data and calculate total usage.
Stakeholder	Customer, Admin

FR04	**Billing**
Description	Based on usage, the system will generate a monthly bill for each customer
Stakeholder	Admin, Customer

FR05	**Payment**
Description	Customers can pay their bills through an online platform or in person at payment centers.
Stakeholder	Customer

FR06	**Excessive Usage Warning**
Description	If a customer uses more water than usual, the system sends a warning to the customer and notifies the admin
Stakeholder	Customer, Admin

FR07	**Report Issues**
Description	Customers can report problems (leaks, no water, poor quality) via the system, which notifies the repair team for quick action.

Stakeholder	Customer, Repair Team

FR08	**Water Quality Monitoring**
Description	The system monitors water quality and alerts the admin if any issues are detected
Stakeholder	Admin

FR09	**Logout**
Description	Users can log out of the system after completing their tasks to ensure security
Stakeholder	Admin, Customer

FR10	**User Profile Management**
Description	Customers can manage and update their profile information
Stakeholder	Customer

Sequence Diagram

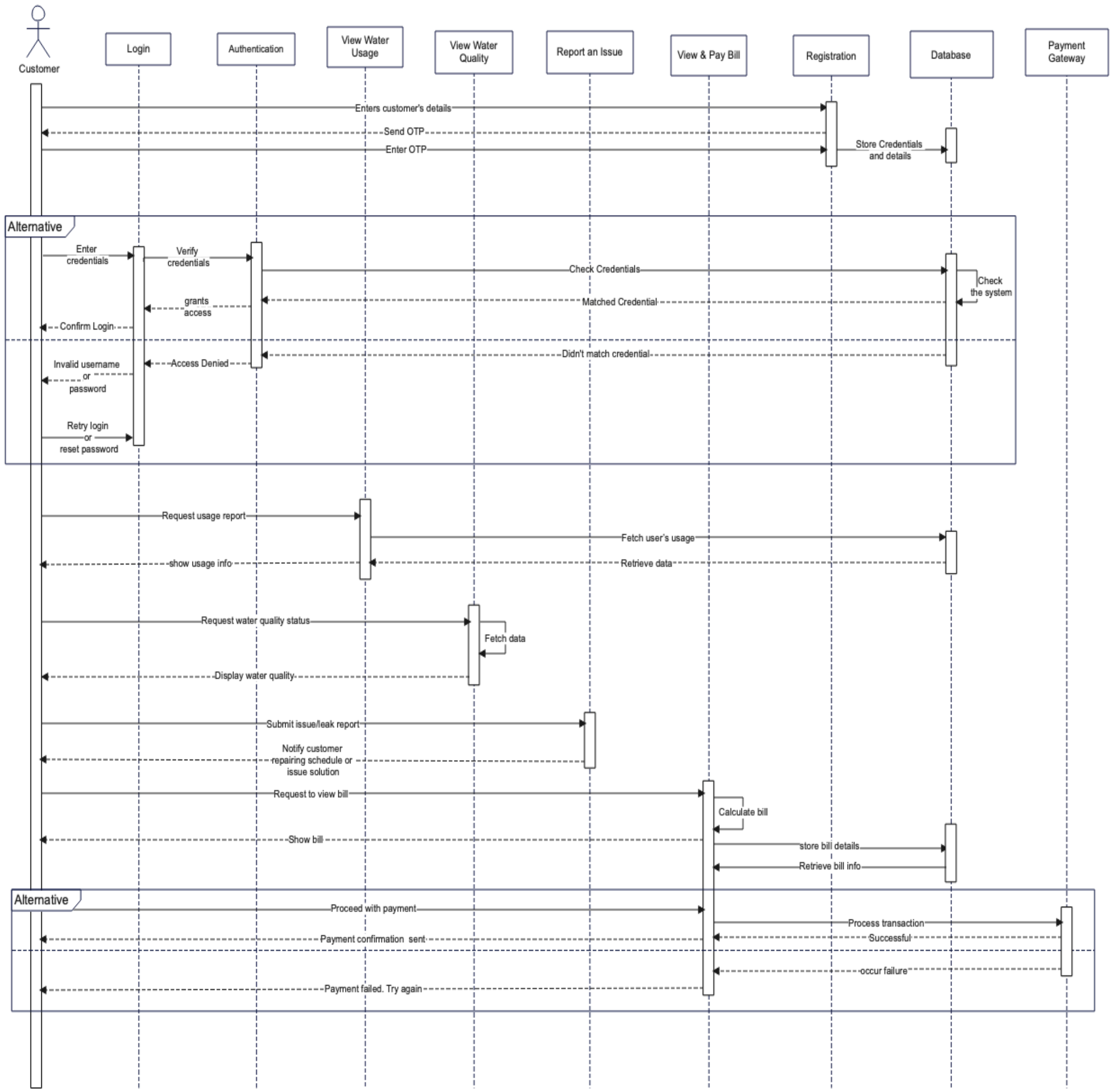


Fig. Customer Sequence Diagram

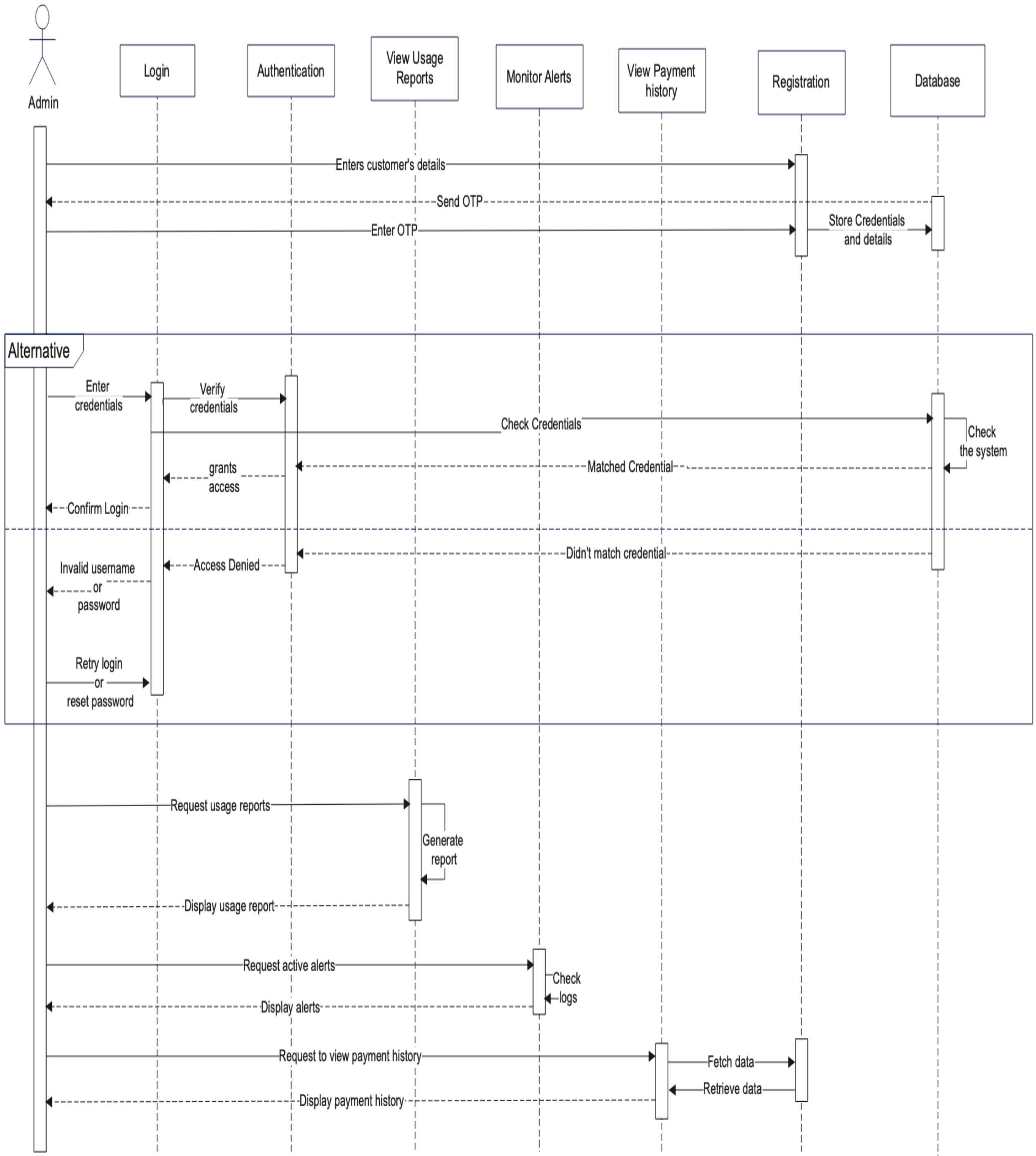


Fig. Admin Sequence Diagram

Data Flow Diagram

Level 0:

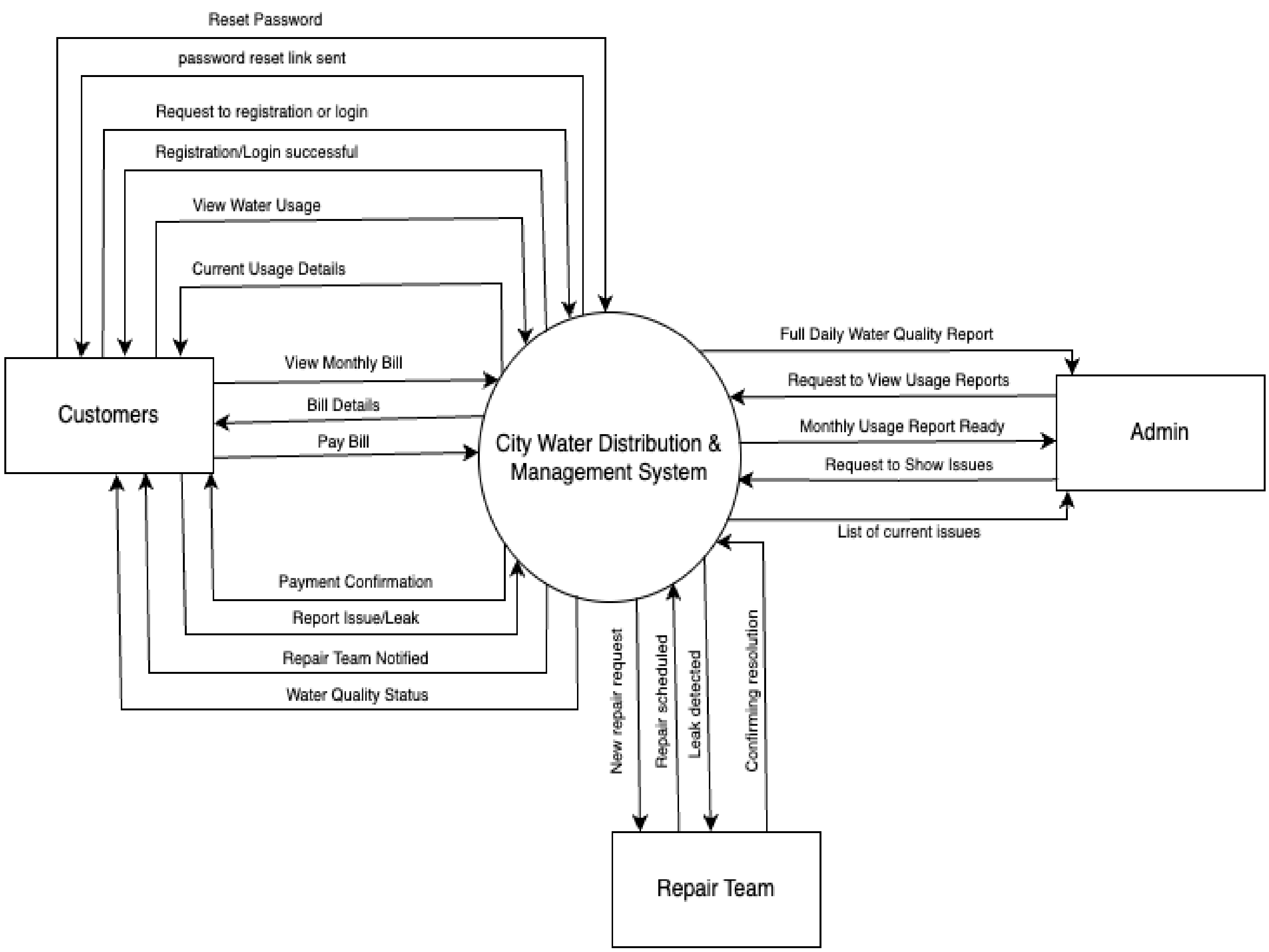


Fig. DFD level 0

Level 1:

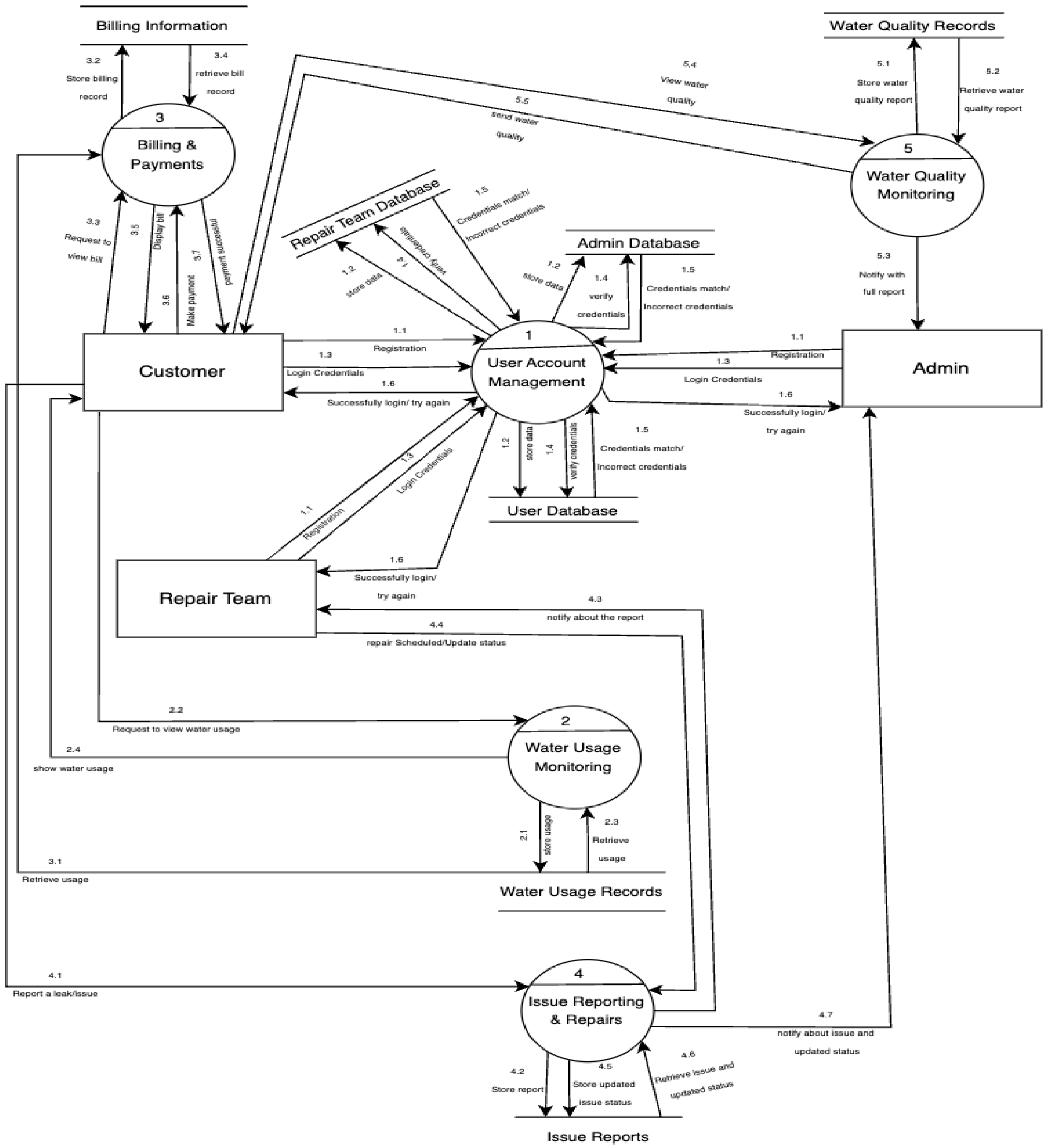


Fig. DFD level 1

Activity Diagram

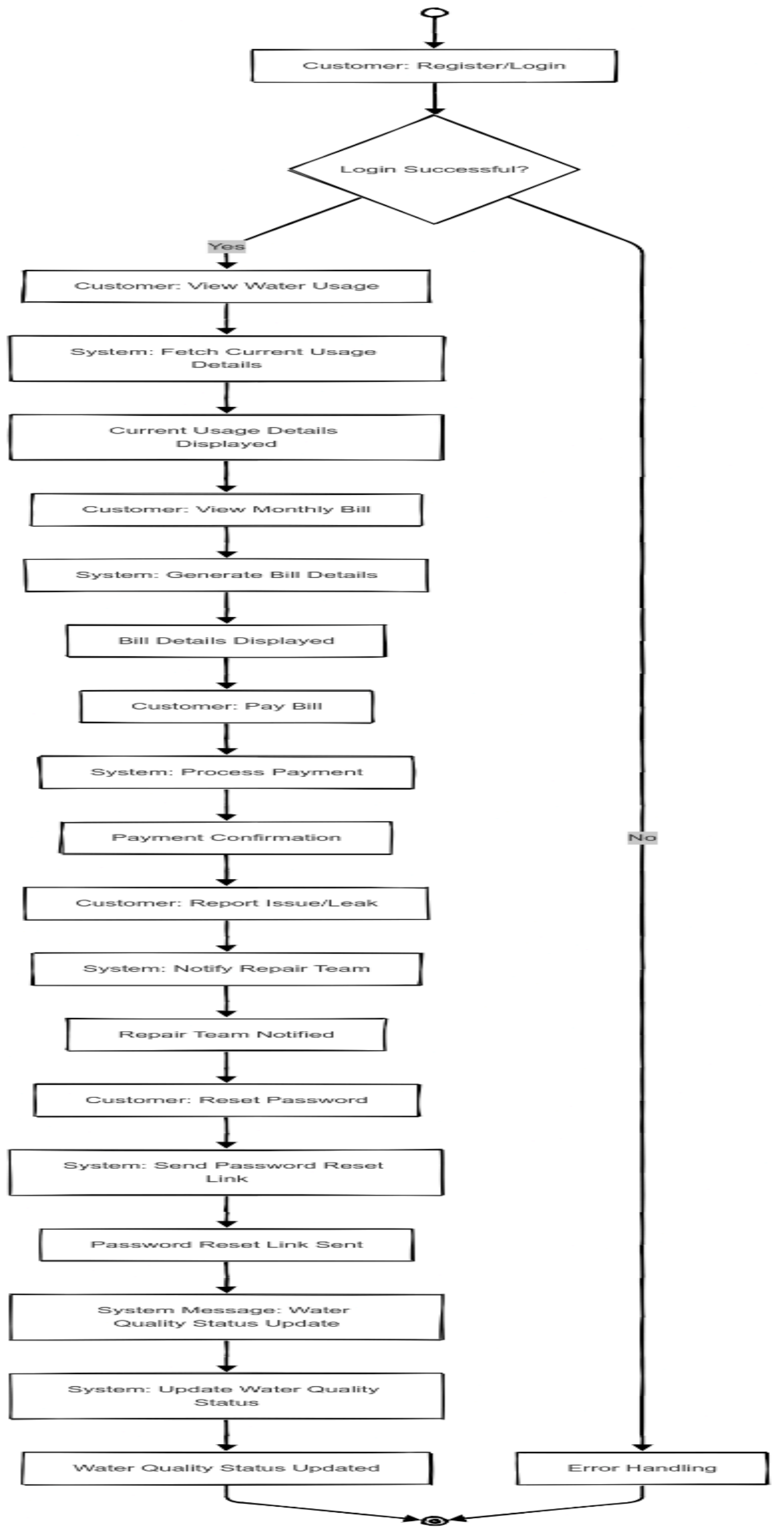


Fig. Customer Activity Diagram

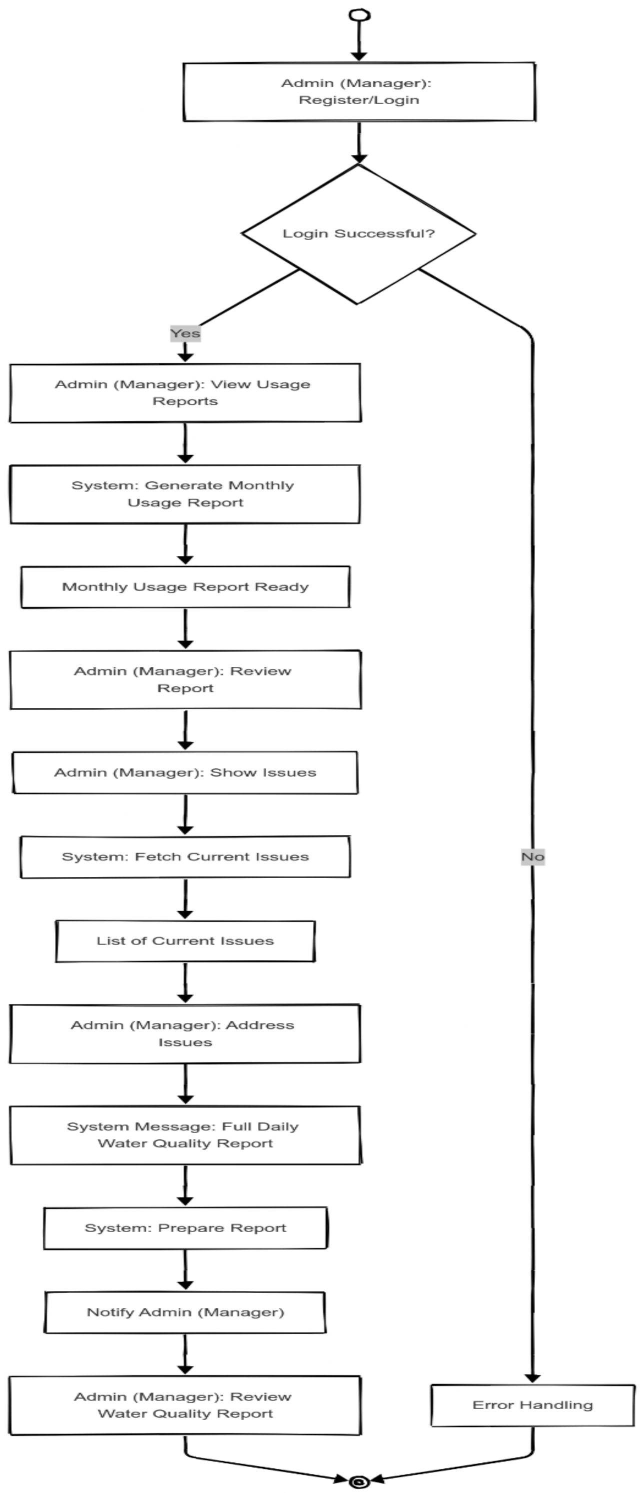
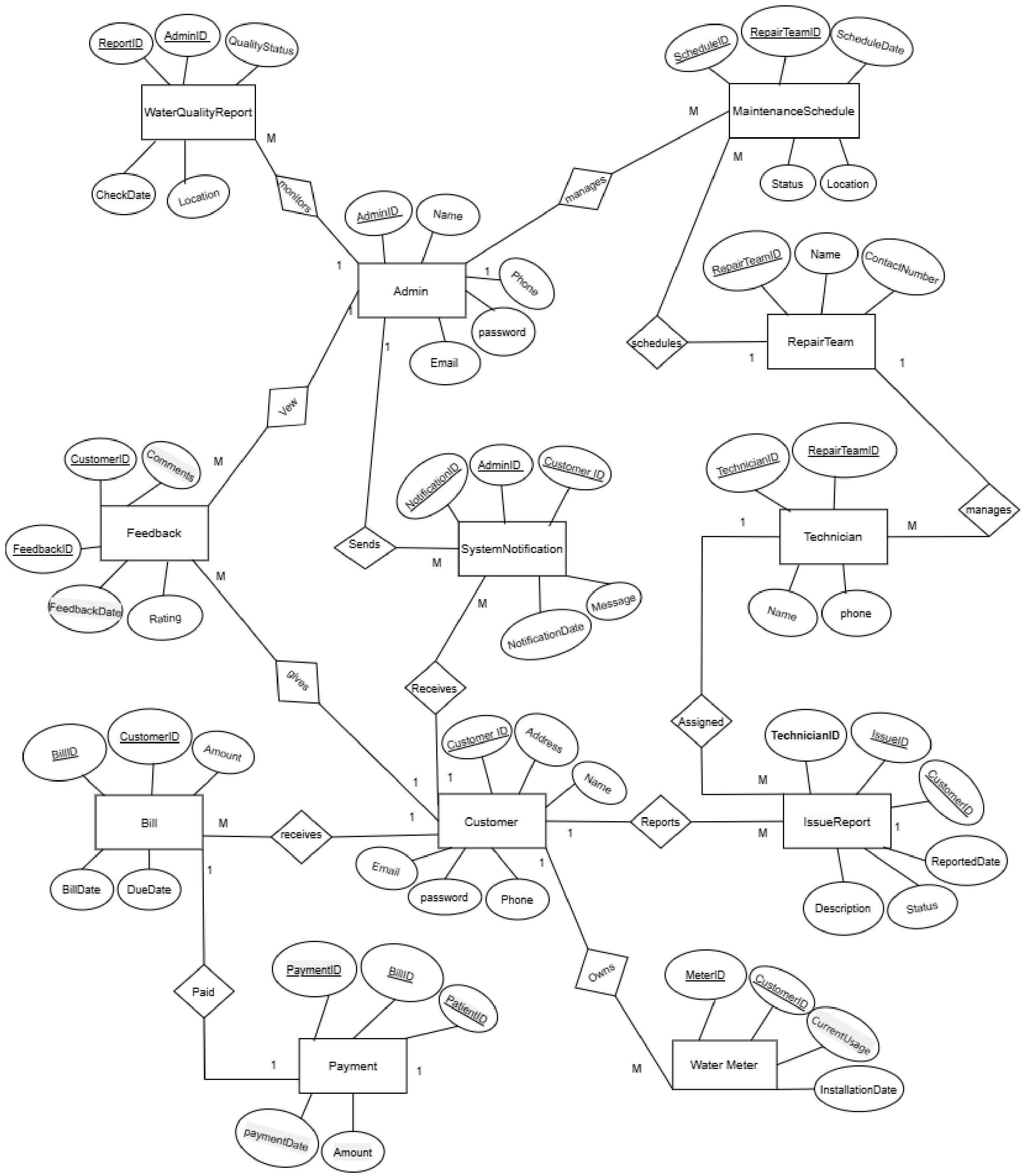


Fig. Admin Activity Diagram

Use Case Diagram

ER Diagram



ReportID
AdminID
QualityStatus
WaterQualityReport
CheckDate
Location
monitors
ScheduleID
RepairTeamID
ScheduleDate
MaintenanceSchedule
Status
Location
manages
AdminID
Name
Admin
Phone
password
Email
RepairTeamID
Name
ContactNumber
RepairTeam
schedules
Vew
CustomerID
Comments
Feedback
FeedbackID
FeedbackDate
Rating
NotificationID
AdminID
Customer ID
SystemNotification
Sends
Message
NotificationDate
TechnicianID
RepairTeamID
Technician
manages
Name
phone
gives
Receives
Assigned
BillID
CustomerID
Amount
Bill
BillDate
DueDate
receives
Customer ID
Address
Name
Customer
Email
password
Phone
Reports
TechnicianID
IssueID
CustomerID
IssueReport
ReportedDate
Description
Status
Paid
PaymentID
BillID
PatientID
Payment
paymentDate
Amount
Owns
MeterID
CustomerID
CurrentUsage
Water Meter
InstallationDate

Class Diagram